Geospatial RAP Tool
Administration and User Guide

1. App Overview

The Geospatial RAP (Reproducible Analytical Pipeline) Tool is a Shiny-based application designed for spatial epidemiological analysis. It generates prevalence maps and spatial clustering outputs (LISA) from aggregated case data.

This tool is part of the RAP ecosystem and operates sequentially:

1. Data Collection: Individual-level suspected case data is collected (e.g., RapidCase Report)
2. Aggregation: Data is aggregated by week and spatial unit using API-POP
3. Spatial Analysis: The aggregated CSV is used as input for this Geospatial RAP Tool

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2. Administrator Setup

This application is designed for simple deployment and does not require database credentials or environment configuration.

- No .Renviron or config.yaml required
- Can run as:
    - Standalone desktop app (Electron)
    - Local or hosted Shiny application

The application can be executed immediately after installation.

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3. Customization and Configuration

3.1 UI Customization (config.css)

The visual appearance of the application is controlled via the file:

www/config.css

Key customizable elements include:

- Backgrounds and layout (navigation bar, sidebar, content area)
- Map background styling
- Input fields and dropdowns

Administrators can modify colors, contrast, and layout by editing the values in this file.

3.2 Geographic Configuration (config.json)

The spatial structure is defined in:

config.json

Each country configuration includes:

- Administrative levels (e.g., level1, level2, level3)
- Mapping between CSV columns and spatial layers
- Spatial identifiers and labels

Example structure:

Argentina → province → department → fraction

This allows the application to support multiple countries without modifying the core code.

3.3 Offline Spatial Data (.rds)

For performance and offline capability, the application uses optimized spatial layers stored as .rds files.

These files are located in the data folder.

To generate them, run the script:

build_data/optimize_spatial_data.R

This process:

- Simplifies geometry
- Standardizes structure
- Improves rendering performance

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4. End-User Guide

Welcome to the Geospatial RAP Tool. This application allows you to explore spatial patterns in epidemiological data through maps and statistical analysis.

Step 1: Prepare Your Data

Ensure your dataset:

- Comes from API-POP
- Is aggregated by spatial unit and epidemiological week
- Is saved as a CSV file

Step 2: Upload and Configure

- Upload your CSV file into the application
- Select:
    - Year
    - Week or range of weeks
    - Geographic level

The application will automatically:

- Detect column names
- Standardize formats
- Prepare the dataset

Step 3: Run the Analysis

The application will:

- Aggregate data for the selected time window
- Calculate prevalence (cases per 100,000 inhabitants)
- Compute:
    - Global Moran’s I
    - Local Moran’s I (LISA clusters)

Step 4: Explore the Results

The application produces:

- Prevalence Map
  A choropleth map showing disease intensity

- LISA Cluster Map
  Identifies spatial clusters:

    - High-High
    - Low-Low
    - High-Low
    - Low-High
    - Not significant

- Results Table
  Tabular output with computed indicators

Step 5: Export Results

Users can download results as CSV files for reporting or further analysis.

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5. Notes and Best Practices

- Geographic identifiers are automatically standardized and padded
- Use appropriate geographic levels:
    - Higher levels for stable trends
    - Lower levels for detailed exploration
- Sparse data may result in non-significant clusters

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6. Troubleshooting

Map appears empty:

- Verify that spatial identifiers match the configuration

No clusters detected:

- Data may not exhibit spatial autocorrelation

Slow performance:

- Ensure optimized .rds layers are being used